

At erection peaks, where vehicles wait for the crane decides whether its queue reaches the whole yard

Cranes as dock resources, the p curve and threshold at each fleet size, crane starvation, the rise and fall with fleet size, and a candidate instability proposition

Dock-mouth blocking at erection peaks: crane utilisation, waiting position and dock-road granularity

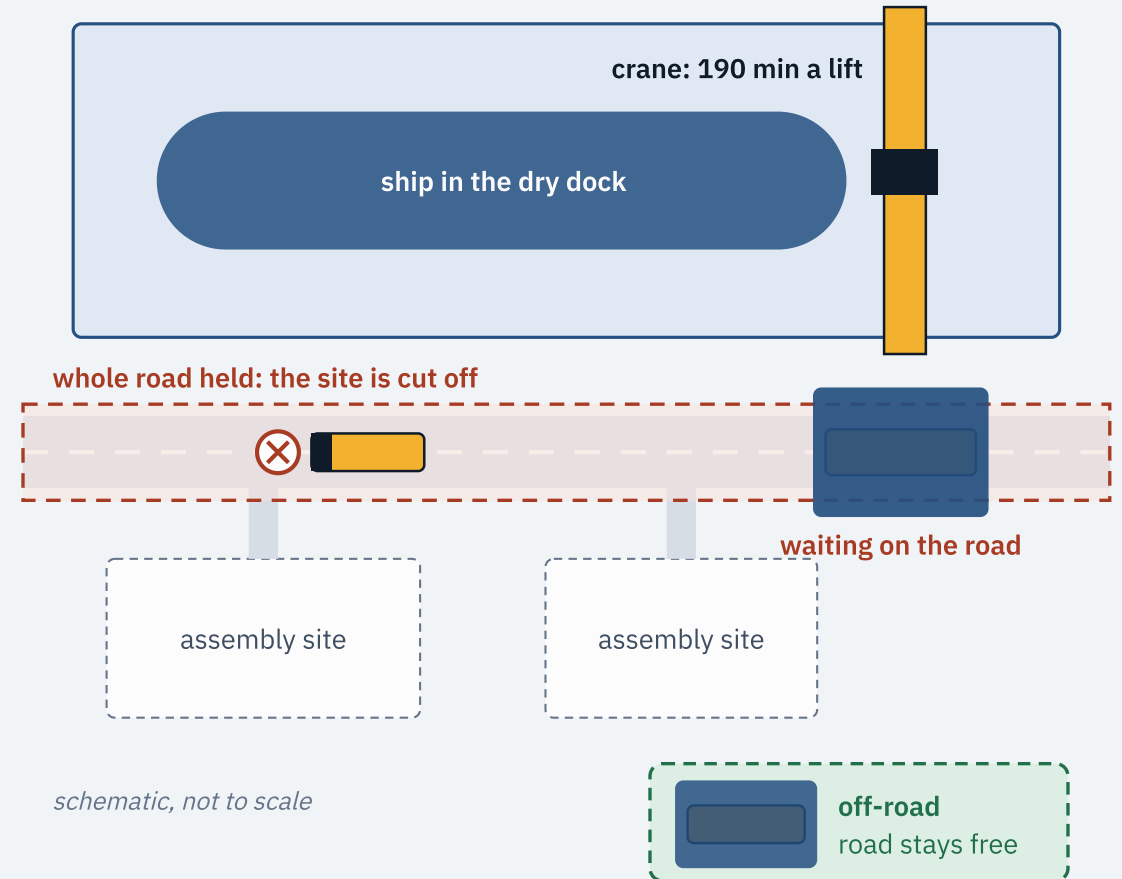
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Basis: this study's simulations and bounds (checked)

This study's claims and hypotheses · decisions of 9 Oct 2026 · literature check

Dock mouth = crane + dock stopping road (in series)



This sub-study in one minute

1

The question

At erection peaks, when does the dock mouth (crane and dock stopping road in series) become the yard's bottleneck? This study's network-bound hypothesis makes it conditional: it depends on whether vehicles waiting for the crane block the way to other stops.

2

What the runs read

Whole-route reservation, waiting off-road, never falls on either yard, at either granularity, for $K = 20-150$ and p up to 0.99 (at most -1.7%). Rules that wait on the road, safe release without teleporting included, all fall with p ; the threshold moves left with K and right with splitting.

3

Effective capacity

At $p = 0.99$ safe release uses only 74–87% of crane-hours, reservation 96–97%: nominal $p < 1$, yet effective capacity falls short. The erection mix's capacity interval [611, 703] becomes [662, 1,241] with the road split.

4

Next

Running: a less conservative deadlock-free rule that re-checks the interval's lower edge and the decline. Then the three-resource network and the block-length curve; floating docks and a finite holding area S . The proposition is being written.

The reading in one sentence: whether a crane queue reaches the whole yard depends on where vehicles wait, how the dock road is split, and how many wait on it.

The order of this briefing (click to jump)

1

Question and position

Hypotheses served: the dock-mouth condition, the decline, vehicle type, the candidate instability proposition

p. 4

2

Cranes and the p curve

Cranes and erection cadence; the p curve at each K ; how the threshold moves with K and splitting; readings for the dock-mouth condition

p. 5

3

Mechanism and effective capacity

Where vehicles wait; crane starvation; teleports counted by task and the split road; the erection mix's capacity interval

p. 9

4

When the decline appears

Safe release rises then falls; the no-clearing switch; three switches and the non-monotonicity theorem

p. 13

5

The proposition and floating docks

Starvation caused by blocking; the three-resource network; floating docks and vehicle type

p. 15

6

Progress, plan and journals

Established, running and queued tasks; target journals (school directory 2026)

p. 18

Question and position: the dock-mouth section of chapter 3

At erection peaks, through what channel does the crane queue become a capacity loss for the whole yard, and can the waiting position and the granularity of the dock road keep it at the dock mouth?

HYPOTHESIS OR PROPOSITION	STATEMENT	THIS STUDY SUPPLIES
Network-bound hypothesis: dock-mouth condition	Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops, that is, where they wait and whether the dock road is used in sections.	ρ curves at each K: three rules $\times$ two road granularities $\times K = 20\text{--}150$
Decline hypothesis	As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S.	Safe release rising then falling and the no-clearing switch; a re-check with a less conservative rule; holding area S; the non-monotonicity theorem
Dock-mouth instability proposition (candidate)	With nominal load $\rho < 1$, starvation caused by blocking destabilises the system; with off-road waiting or a dock road used in sections, stability is restored.	The crane starvation rate (read); the three-resource network (planned)
Vehicle-type hypothesis: case	Multi-transporter carriage of giant blocks for floating docks.	A floating-dock scenario (quay-side sink + floating crane, planned)

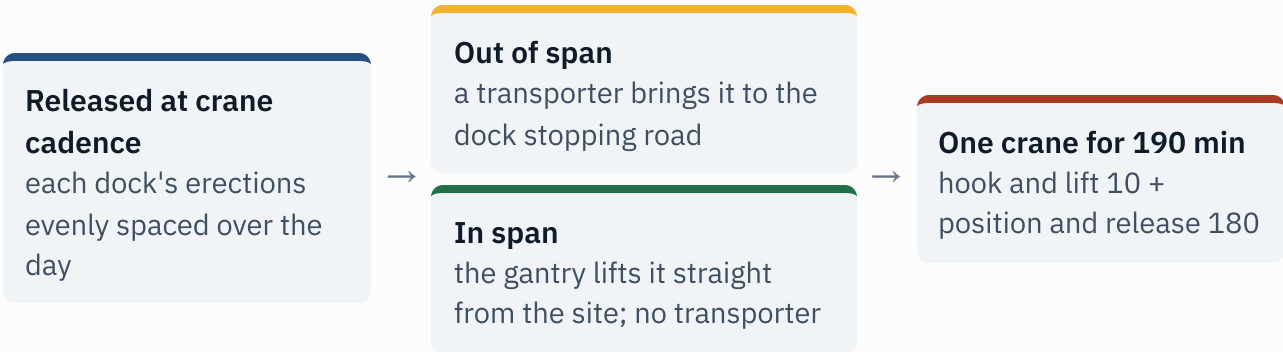
Related claims: the trapezoid law's falling branch appears “when waiting crowds the road”; traffic ordering includes waiting points and block granularity. Each conclusion states its conditions once: traffic rule, task-volume basis, resource granularity, yard and map version.

Sources: this study’s claims (the trapezoid law; from dispatching to traffic ordering), hypotheses (network bound, decline, vehicle type), the non-monotonicity theorem, the candidate instability proposition, and its wording rules. The English and Korean statements are translated from the Chinese, which governs.

Cranes are dock resources: erections are released at crane cadence, and in-span lifts take the crane but not the road

YARD · DOCK	CRANES COUNTED (CAPACITY ≥ 100 T)	NO.
Okpo · dock 1	900 t gantry, 200 t jib crane	2
Okpo · dock 2	900 t gantry	1
Yantai · dock 1	20,000 t bridge crane (Taishan), 370 t gantry	2
Yantai · dock 2	370 t gantry	1

Jib and tower cranes below 100 t (the mean block weight) are not counted; assigning Yantai's two 370 t gantries one per dock is a scenario choice. Okpo gantry spans: 206 m on dock 1 (Riviera 2014), 148 m on dock 2 (The Motorship).



$$\rho = \text{erections/day} \times 190 \text{ min} \div (n_{\text{crane}} \times 960 \text{ min})$$

Three cranes erect at most $3 \times 960 \div 190 = 15.2$ a day

0.88

(0.78-0.97)

Okpo: erections = $2 \times 20 \text{ ships} \times 100 \div 300 = 13.3$ a day (range 11.9–14.7). 20 = LNG carriers delivered in 2025 (estimated from IR), 100 = midpoint of DSME's patent “usually 90–110”, 2 = peak day. **A scenario reading, kept out of the network-bound hypothesis** (9 Oct 2026).

0.13

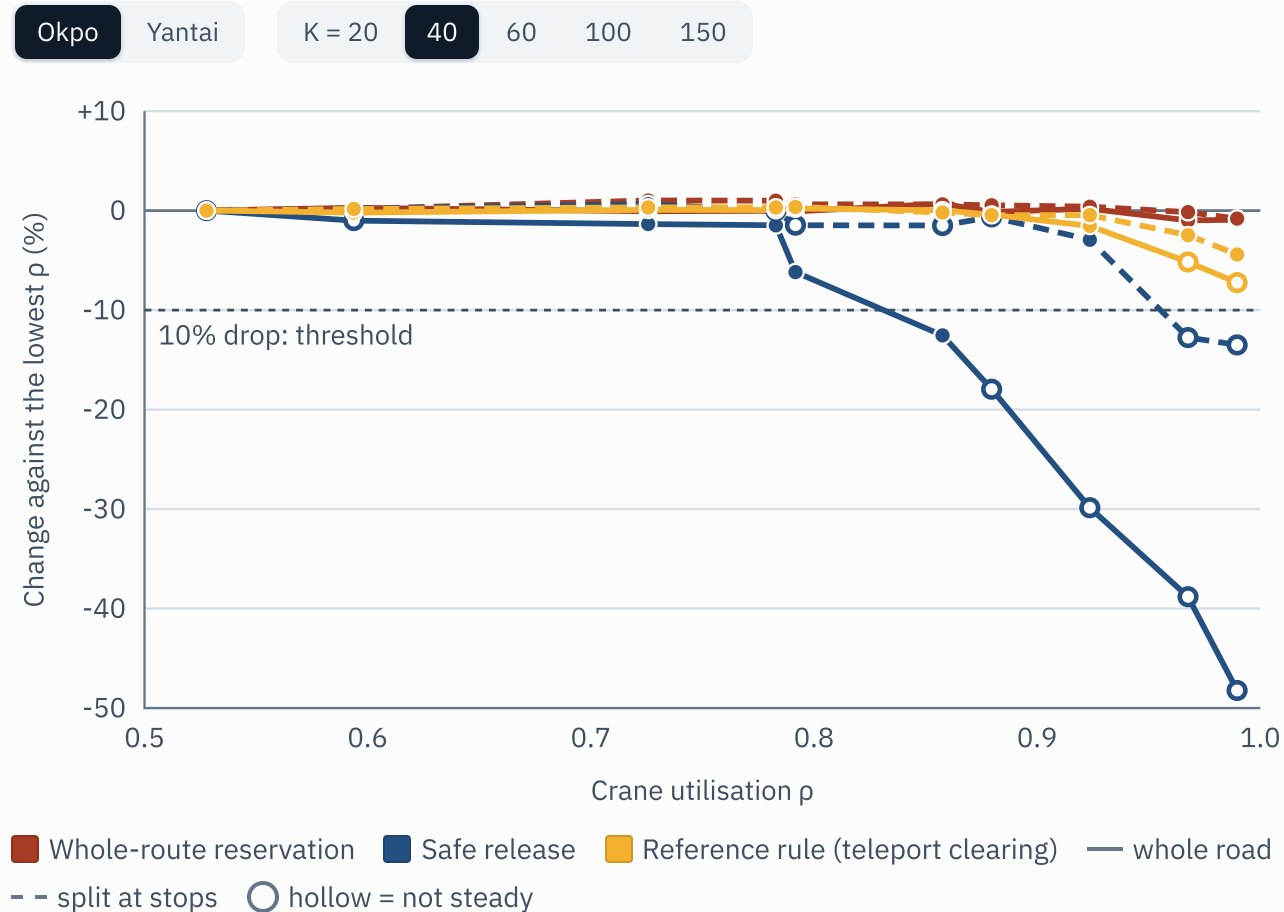
Yantai: 2 erections a day (public output); cranes far from tight.

12.5%

Okpo's in-span share: one assembly site in each dock's gantry span (map cover ≥ 0.5); none at Yantai.

Sources: crane settings and erection volumes of this study (crane lists: Hanwha Ocean Facilities page; CIMC Raffles 2012 EIA notice; Hoist Magazine 2008; in-span shares computed on the Okpo map). Crane cycle from Zhang 2025 (another yard); units per ship from DSME patent KR20090050364A and Roh & Lee 2006, p. 116 (89); ships a year estimated from Hanwha Ocean IR.

The ρ curve at each K: reservation (off-road) does not fall; on-road rules, safe release without teleports included, all fall with ρ



-0.9% / -0.8%

Reservation (waiting off-road): change at $K = 40$ and $p = 0.99$ (whole / split)

0.83 → 0.96*

Safe release (no teleporting), threshold at $K = 40$ (change at $p = 0.99$ if under 10%): whole → split

-7.2% → -4.4%

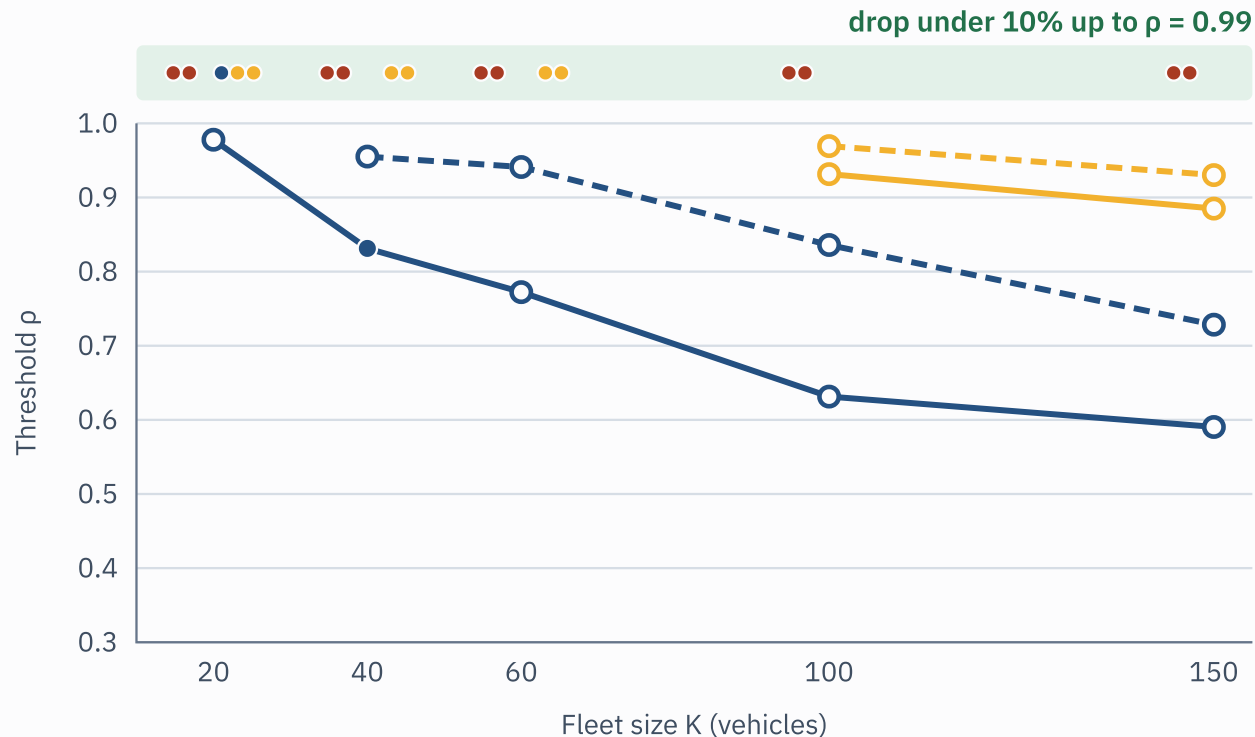
Reference rule (teleport clearing), threshold at $K = 40$ (change at $p = 0.99$ if under 10%): whole → split

Threshold = the p at which the drop first reaches 10%; * = it uses non-steady points. Safe release is already non-steady at the lowest p ; its slide at $K = 100-150$ comes from the criterion itself, not from the dock.

Sources: simulations in this study: the crane-cadence p levels, three rules $\times$ two road granularities, saturated demand, 16 h working day, 8 days (day 1 warm-up), 10 seeds; change against the yard's lowest p (Okpo 0.53, Yantai 0.13); hollow = fails the steady-state test (mix shift $\leq 2\%$, daily trend $\leq 5\%$); reservation and the reference rule on the whole road reuse the gantry-span runs (reproduced bit for bit). Maps: the Okpo map before floating docks and branch roads, the Yantai map; the rule comparison and retrofit simulations that follow use the Okpo map with them.

The threshold moves left with fleet size and right with splitting; reservation never falls: no single threshold fits every rule and K

Okpo Yantai



0.98* → 0.59*

Safe release, whole road: threshold from K = 20 to K = 150

0.83 → 0.96*

At K = 40, splitting the dock road at its stops moves the safe-release threshold right (whole → split)

0.89* → 0.93*

Reference rule (teleport clearing), K = 150: whole → split

-1.7%

Reservation: largest drop at $p = 0.99$ over both granularities and every K = 20–150

■ Whole-route reservation
 ■ Safe release
 ■ Reference rule (teleport clearing)
 — whole road
 - - split at stops
 ○ hollow = not steady

Sources: simulations in this study. Threshold = the p at which throughput first falls 10% below the lowest p (linear between neighbouring levels); hollow and * = the points behind the threshold include non-steady ones; green band at the top = the drop stays under 10% up to $p = 0.99$ (hover for the change at $p = 0.99$). Conditions as on the previous slide.

The dock-mouth condition: readings name the rule and K; wording and main basis follow the rule comparison

Dock-mouth condition: whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops, that is, where they wait and whether the dock road is used in sections.

Current condition note (9 Oct 2026): the old thresholds (Okpo about 0.85, Yantai about 0.6–0.7) were read only under the reference rule (teleport clearing) with the dock road as one resource. The physical-rule runs show the threshold moves with rule, granularity and K; the table gives every cell.

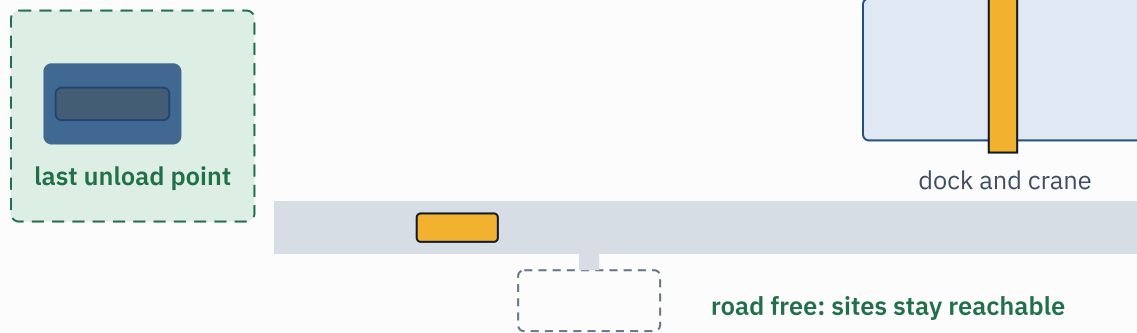
RULE · GRANULARITY	OKPO					YANTAI				
	K = 20	K = 40	K = 60	K = 100	K = 150	K = 20	K = 40	K = 60	K = 100	K = 150
■ Whole-route reservation · whole road	– (–1.2)	– (–0.9)	– (–0.8)	– (–1.7)	– (–0.6)	– (+0.5)	– (+1.8)	– (+0.9)	– (+1.2)	– (+1.4)
■ Whole-route reservation · split at stops	– (–1.6)	– (–0.8)	– (–1.2)	– (–0.2)	– (–0.8)	– (+1.4)	– (+0.8)	– (+1.8)	– (+0.7)	– (+1.8)
■ Safe release · whole road	0.98*	0.83	0.77*	0.63*	0.59*	0.93*	0.70*	0.56	0.39*	0.38*
■ Safe release · split at stops	– (–4.5)	0.96*	0.94*	0.84*	0.73*	0.97*	0.80*	0.70	0.55*	0.43*
■ Reference rule (teleport clearing) · whole road	– (–3.7)	– (–7.2)	– (–9.8)	0.93*	0.89*	0.98*	0.90*	0.85*	0.79*	0.66
■ Reference rule (teleport clearing) · split at stops	– (–2.2)	– (–4.4)	– (–6.4)	0.97*	0.93*	– (–5.7)	0.94*	0.88*	0.80*	0.74*

After the rule comparison: the dock-mouth condition's wording (one option: a qualitative layer plus a numeric layer naming rule and K); the network-bound hypothesis's new vehicle numbers; the main basis (16 h or 24 h day, daily mean or peak). If the best deadlock-free rule changes, the safe-release rows gain a less conservative companion.

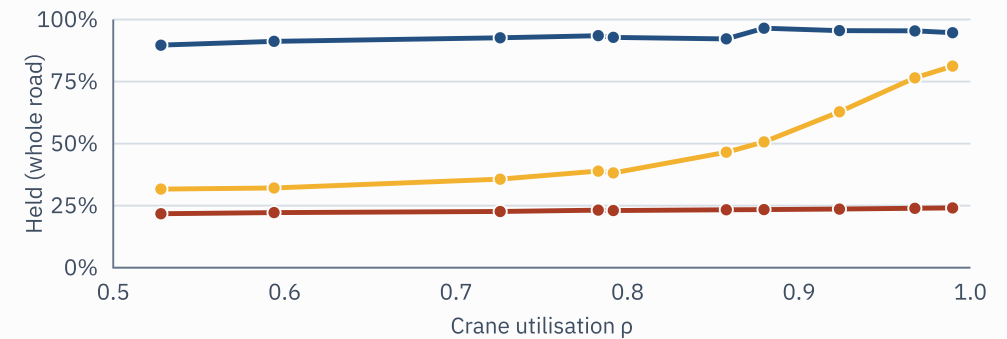
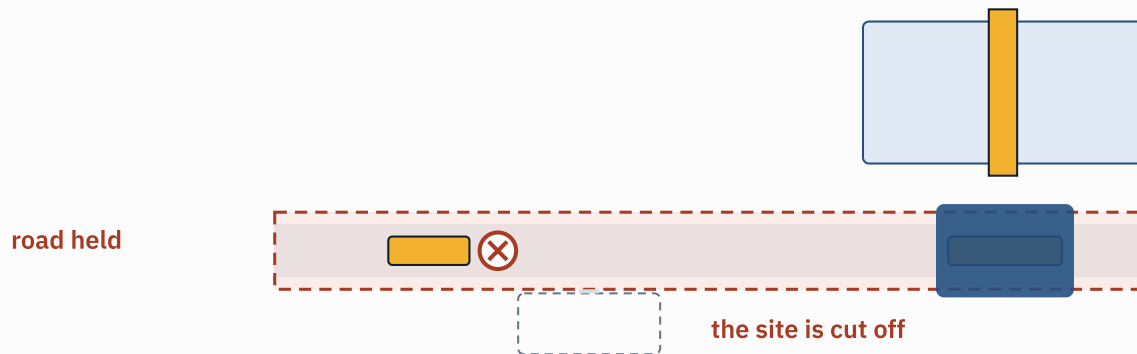
Sources: this study's network-bound hypothesis (quoted); simulations in this study. Bold = threshold p; * = uses non-steady points; – (x) = drop under 10% up to p = 0.99, change at p = 0.99 (%) in brackets. Conditions: saturated demand, 16 h working day, 8 days, 10 seeds, the Okpo map before floating docks and branch roads, the Yantai map.

The mechanism: the difference is where vehicles wait – at their last unloading point (off-road) or on the dock road

Reservation: crane slot booked with the route



Waiting on the road (safe release, reference rule)



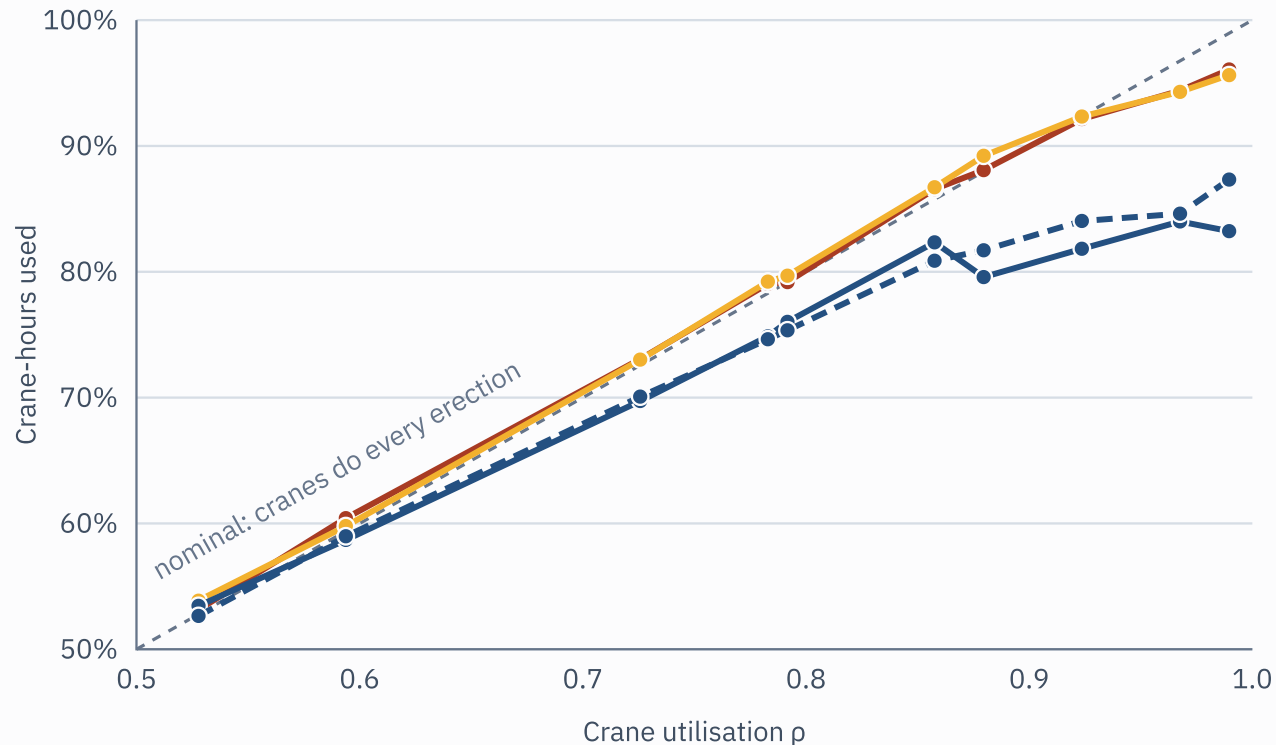
■ Whole-route reservation ■ Safe release ■ Reference rule (teleport clearing)

- **The cost moves off-road:** at $p = 0.88$ and $K = 150$ the dock stopping road is held only about 23% of the time under reservation (vehicles wait at their last unloading point); about 51% under the reference rule and about 97% under safe release (refused vehicles wait on their current road)
- **Split road:** a vehicle waiting for the crane holds only the 50 m at the dock; under the reference rule at $p = 0.99$, held time falls from 81% for the whole dock-1 road to 65%
- **In-span lifts** (same p): waiting at the dock falls by about a quarter, the plateau does not change: p sets the strength, the in-span share only changes how many vehicles queue on the road

Crane starvation: nominal $\rho < 1$, yet effective capacity falls short – at $\rho = 0.99$ safe release uses only 74–87% of crane-hours

Okpo

Yantai



Whole-route reservation Reference rule (teleport clearing) Safe release — whole road

- - split at stops

83% / 87%

Safe release at $\rho = 0.99$, crane-hours used (whole / split): 12–16 points below nominal; erections pile up and the system is unstable

96%

Reservation: cranes close to nominal; vehicles wait at stops for their slot, nothing reaches the network

96%

Reference rule: the cranes lose almost nothing; the plateau falls because the dock queue holds the dock-1 road (81% of the time) and blocks other flows

Starvation = share of time a crane is free while a released erection has not started; under reservation at high ρ about half of the “stuck” time is waiting for the crane slot ($\rho = 0.99$: 53–58%).

Sources: simulations in this study: saturated demand, $K = 150$, 16 h working day, 8 days, 10 seeds; crane-hours used = time the cranes are held ÷ time available, dashed diagonal = nominal ρ ; starvation includes the normal lead time of fetching an erection, with the part stuck in the network reported separately (on hover). The Okpo map before floating docks and branch roads, the Yantai map.

40–50% of loaded erections were teleported, so the result rests on reservation and safe release; the split road is now built in

① Teleports counted by task (reference rule, whole road, K = 150)

36–47%

Okpo: loaded erections teleported at least once

39–49%

Yantai: the same

39–56%

of these, teleported twice or more (both yards, about half)

7–11

most teleports of a single task

By construction a vehicle waiting for the crane or unloading at the dock edge is never cleared; those teleported are carrying an erection to the dock, or hold the dock stopping road while waiting for the next section, and each leaves the network for 900 s.

So the reference rule (teleport clearing) is a comparison only; the dock-mouth result rests on whole-route reservation and safe release, neither of which teleports.

② Okpo's dock-1 road: 1,095 m long, 9 m wide, three stops along it



One resource (comparison): one vehicle at the dock holds all 1,095 m



Split at stops: bounds 25 m either side, 7 sections; only the 50 m at the dock is held

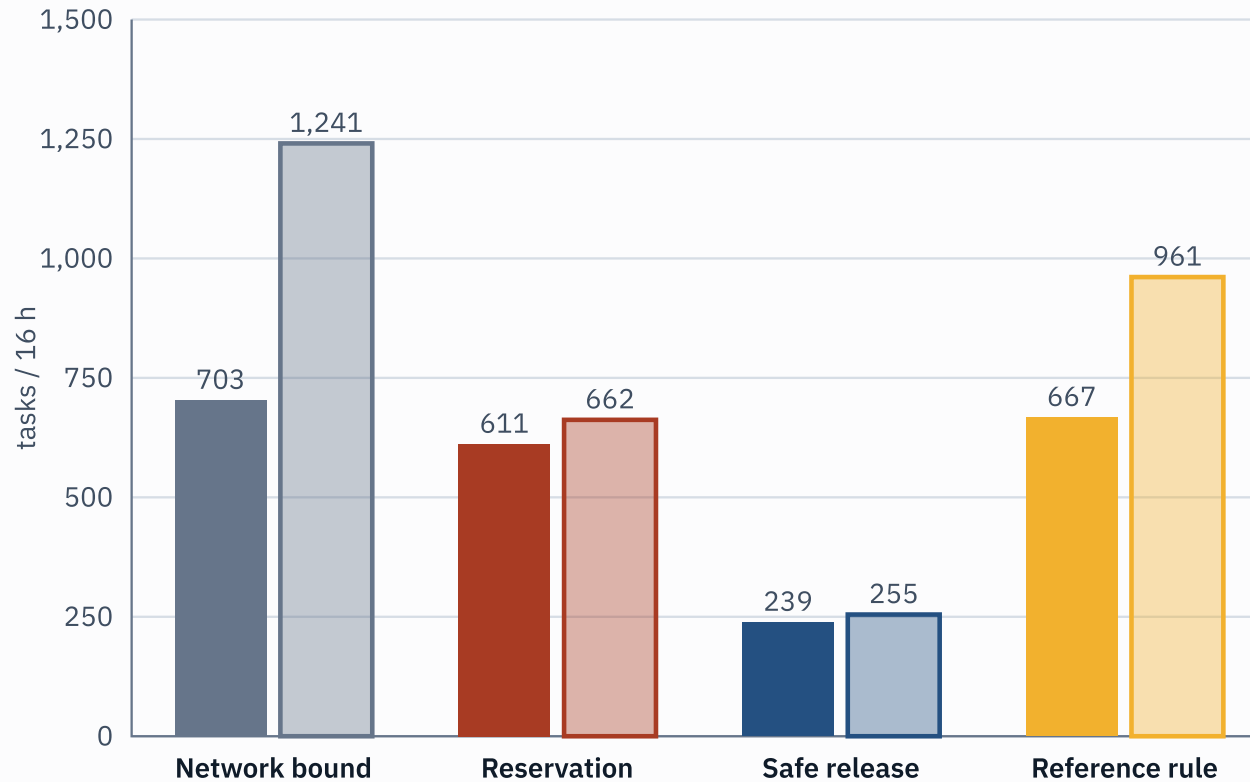
0 m

1,095 m

The split is now a network feature (the road split at its stops): roads over 200 m with roadside stops get bounds 25 m either side of each stop. One rule for both yards picks exactly the 4 dock roads; 200 m and 25 m are scenario settings borrowed from railway block sections (UIC 2013) and single-lane mine roadways (Eustace et al. 2021).

Imagery check (9 Oct 2026): while a vehicle unloads or waits at the dock edge, neighbouring assembly sites **stay reachable**, so the split is closer to Okpo; the whole road is the comparison.

Erection mix: splitting the dock road moves the interval from [611, 703] to [662, 1,241] and the bottleneck from road to rule



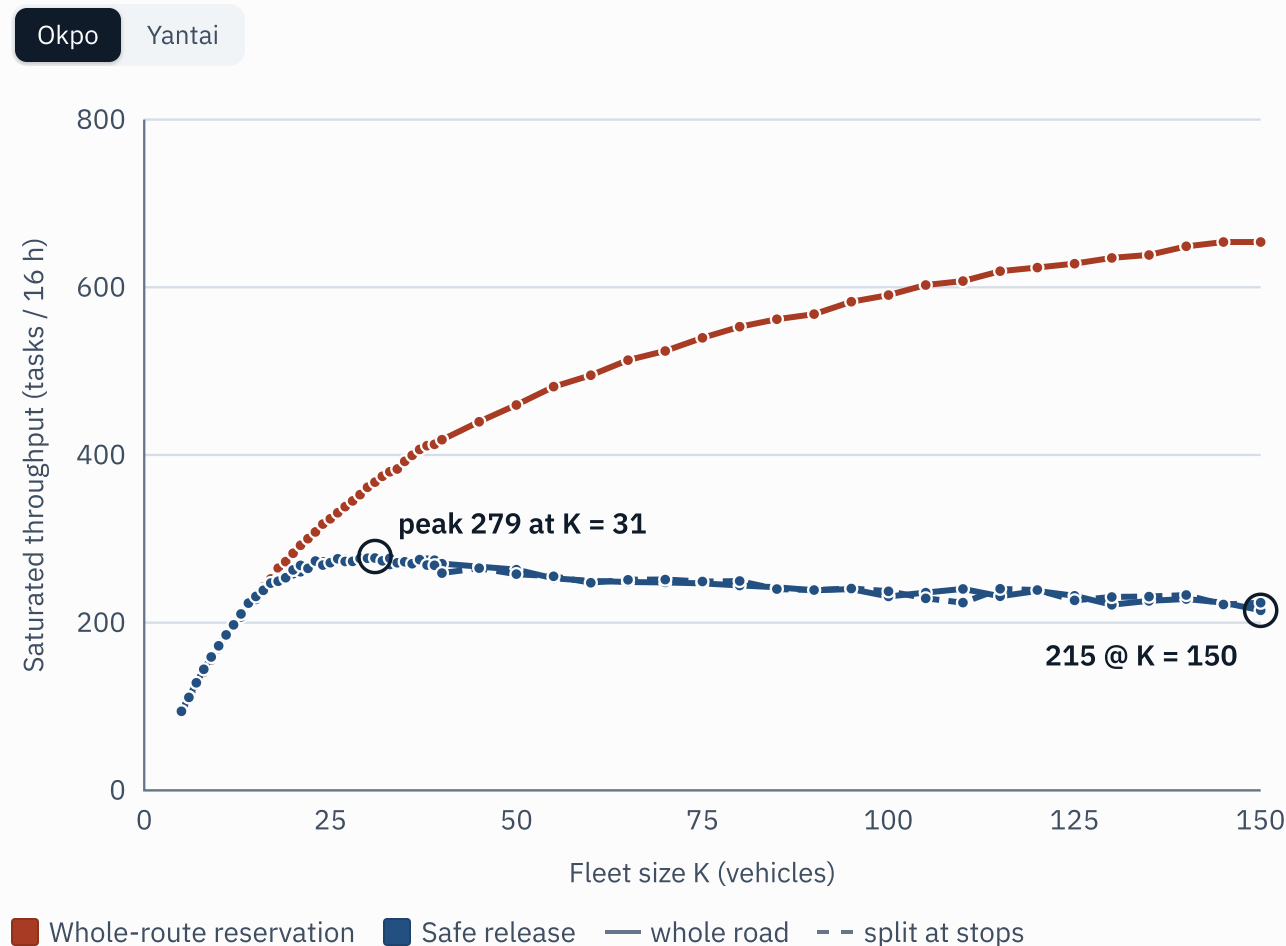
■ whole road □ split at stops

- Erection mix: [611, 703] → split [662, 1,241]; upper edge +76.5%, lower edge +8.3%
 - Plateau / network bound: erection mix 0.87 → split 0.53, main scenario 0.34–0.36: with the split the bottleneck moves from road to rule (ordering vs. exchange rate)
 - Okpo main scenario: [627, 1,720] → split [630, 1,857], upper edge only (+8.0%; lower +0.3%); Yantai [455, 1,252]
 - **System bound:** 697 for the erection mix (no sampling noise); reservation reaches 86% of it on the completed mix, so road changes still pay
- Interval = [best deadlock-free rule, network bound]; the best tested is whole-route reservation (above safe release for every $K \geq 20$). The split-road reference rule has direction locks, but its gain still includes teleports (2,394 deadlocks a day at $K = 150$): comparison only.

A 3-seed probe (9 Oct 2026) saw it first (bound 703 → 1,241, reservation +10%); 10 seeds and the full K grid confirm it (reservation 585 → 627 at $K = 100$, +7%).

Sources: simulations in this study: erection mix = erections at 15%, no crane limit; saturated, $K = 5-150$, 4 days, 10 seeds; plateau = mean over $K = 100-150$; 16 h working day; the Okpo map before floating docks and branch roads; reservation and the reference rule on the whole road reuse the earlier bound and crane-cadence runs; the system bound comes from the bound computation. The main scenario's lower edge 627 is the unrounded 627.45.

Decline hypothesis: safe release (no teleports, no deadlock) rises then falls; reservation, waiting off-road, rises to a plateau



Decline hypothesis: with off-road waiting and a deadlock-free rule, throughput never falls as vehicles are added: a plateau. As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S .

279 → 215

Safe release (whole road): peak at $K = 31 \rightarrow K = 150$ (–23%)

627

Reservation: rises monotonically to its plateau (mean over $K = 100$ – 150 ; 654 at $K = 150$)

984

Plateau of the reference rule (teleport clearing, off the chart); highest 1,028 at $K = 150$, with no fall like safe release on the same network

The peak and the fall are a property of the safe-release criterion: about 2/3 of its waits start with a refusal although capacity is free. Without clearing the network jams, sooner with more K (Okpo: 4.3 h at $K = 5$, under 1 h for $K \geq 40$). Next: a looser rule re-checks the fall; then a holding area S .

Sources: this study's decline hypothesis (quoted); simulations in this study: main scenario (the fleet-sizing paper's task-flow mix, Roh & Cha 2011), saturated demand, 16 h working day, 4 days, mean of 10 seeds; reservation and the reference rule reuse the earlier bound and crane-cadence runs; the Okpo map before floating docks and branch roads, the Yantai map.

The decline's switches: clearing and waiting swept, holding area S next; the theorem is proved on a corridor first

① Holding area S

idle and waiting vehicles beyond S occupy the roadside; read $K_0(S)$ planned

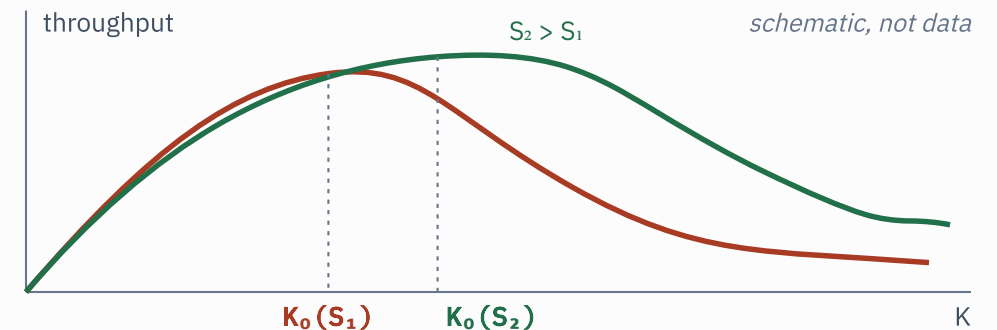
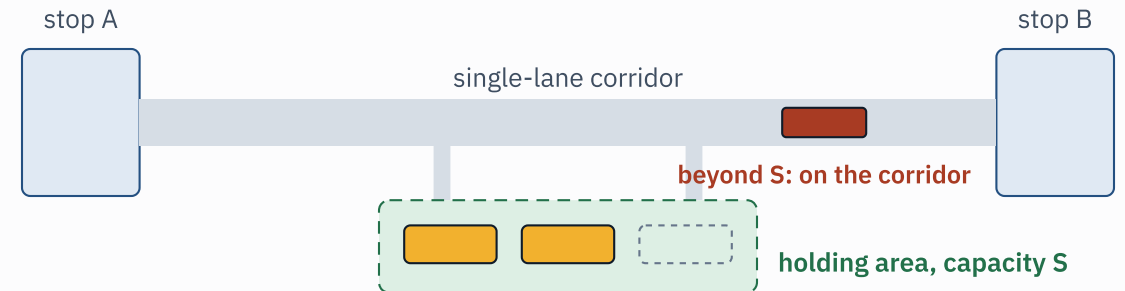
② Clearing mode

safe release (banker's-algorithm check, no vehicle leaves the network): rises then falls; no clearing: the network jams, sooner with more K ; teleporting (reference): monotone to a plateau read

③ Where to wait for the crane

off-road: no fall; on the dock road: falls with p , threshold moving left with K ; split: moves right. holding bays and block length: retrofit runs read · planned

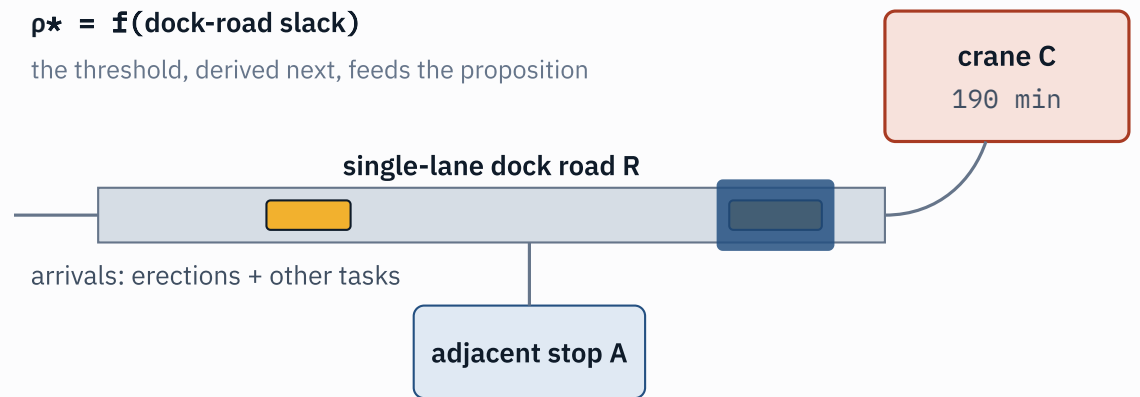
If it holds: the conditions under which more vehicles hurt, and “how many vehicles a holding area is worth”. **If not:** a redundancy index, the ratio of marginal throughput to the free-flow margin. Good orchestration can push K_0 back: a looser deadlock-free rule re-checks the fall next.



Non-monotonicity theorem: on a single-lane corridor with a holding area of capacity S , under greedy release, throughput falls strictly for $K > K_0(S)$. Method: a sample-path construction. Output: chapter 5's design rule “how many vehicles a holding area is worth”.

Candidate dock-mouth instability proposition: at nominal load $\rho < 1$, blocking-induced starvation can destabilise the system – read without teleporting

- **The channel:** vehicles waiting for the crane block the way to other stops $\rightarrow$ later erections cannot reach the dock and the crane idles because upstream is blocked $\rightarrow$ effective capacity falls below 1, and the system is unstable although nominal $\rho < 1$
- **Restoring stability:** waiting off-road (reservation, a holding bay at the dock) or a dock road used in sections $\rightarrow$ the stable range returns to $\rho < 1$
- **Observable:** the crane starvation rate (already read: at $\rho = 0.99$ safe release uses only 74–87% of crane-hours, reservation 96–97%; on Yantai under the reference rule, starvation stuck in the network is twice reservation's)



Wording: we extend the blocking-induced starvation of queueing networks with blocking to a dock mouth where a crane is in series with an exclusively occupied road. Status: simulations under rules without teleporting support the mechanism; the proposition is being written, and the three-resource network gives the threshold ρ^* .

Theory and position: transferred from queueing networks with blocking; the closest work in the field does not narrow the claim

THEORY SOURCE	WORKS	USE
Queueing networks with blocking	Onvural 1990; Balsamo et al. 2001 checked	foundation
Policy-induced instability at nominal load < 1	Kumar & Seidman 1990; Lu & Kumar 1991; Rybko & Stolyar 1992; Dai 1995 checked	analogy, transfer source; fluid models as the proof tool
Deadlock avoidance (banker's algorithm)	Habermann 1969 to verify ; Reveliotis 2000	source of safe release; conservative on single lanes in general

“Nominal load < 1 at each station is not sufficient for stability” is best quoted from Dai 1995's abstract; Dai's model has no blocking and infinite buffers, a difference stated when transferring it to road occupancy.

	TAO ET AL. 2017	THIS STUDY
Road occupancy	capacity counted in vehicles (2 on main roads, 1 on side roads)	exclusive use and blocking time that change with the load
Rules, deadlock	one (first come, first through), deadlock not handled	three rules; reservation and safe release are deadlock-free
Road-level output	none (no count, duration or road utilisation of blocking reported)	thresholds, dock road held, crane starvation
Bound	none (simulation only)	network and route-free bounds, capacity interval
Yards	one	two (more in chapter 6)

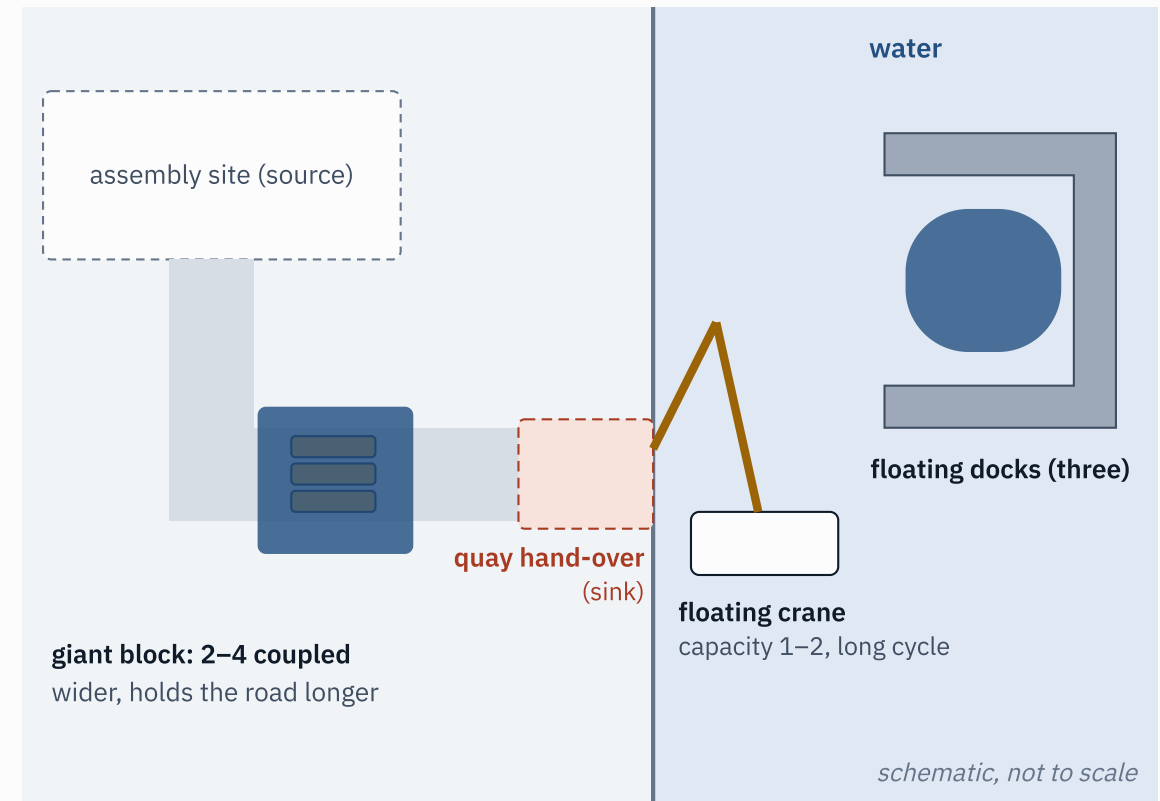
What is new: Zhang 2025 schedules gantry cranes and transporters separately; here crane cadence and road-owning loaded vehicles share one simulation and one bound. Within the searches of this study's literature check, this is the first work to read shipyard throughput with waiting position, control granularity, fleet size and crane utilisation as joint variables.

Sources: this study’s literature check: item-by-item comparison with the full text of Tao et al. 2017; bibliographic checks and quotations of transfer sources; comparison with Zhang 2025. Transfer sources serve as theoretical support, not as prior claims.

Floating docks: a quay-side sink plus a floating crane; giant blocks on coupled transporters are a vehicle-type case

- **Three floating docks:** Hanwha Ocean IR 4Q24 p. 29 and 2Q26 p. 27, “Floating Dock : 3EA”; on the imagery, dock 1 $\approx 457 \times 82$ m, 2 $\approx 360 \times 77$ m, 3 $\approx 472 \times 80$ m, container ships in 1 and 3
- **Fits the base scenario:** LNG carriers in dry docks 1 and 2 (confirmed; four at once in dock 1, Seoul Shinmun 2024, Hanwha 2025); all 33 merchant ships there would give $p = 1.29\text{--}1.60$
- **Model:** Okpo lifts with floating cranes (3,600 t class, two in tandem for big blocks): vehicles bring giant blocks to a shore site in reach (the sink); the floating crane is a resource of capacity 1–2 with a long cycle, reusing the crane code; roll-on only as comparison
- **Giant blocks:** over 500 t, sometimes 4 coupled transporters; “about 10 giant blocks a ship” is from 고시근 et al. 2011 and Samsung, not Okpo; wider, holding the road longer, returning as a team $\rightarrow$ vehicle-type hypothesis; the fleet-sizing paper's coupled carriage

Floating docks follow the rule comparison and retrofits; the Okpo map with floating docks and branch roads comes first. Each dock's size and the speed of coupled transporters have no public source yet: scenario settings, reported side by side.



Established (all checked): each feeds a thesis figure or section

ITEM	STATUS	OUTPUT (WHERE IN THE THESIS)
Cranes as dock resources, erections at crane cadence, in-span lifts; the crane row and window edge term of the network bound (draft proof exists; being rewritten)	established	Chapter 3 methods and parameter table
Physical rules: safe release (exact, deadlock-free), no-clearing switch; split road; crane starvation, teleport diagnosis; ρ curve and threshold at each K ; capacity intervals; the decline's rise and fall	established	Chapter 3: dock-section mechanism figure and threshold table; interval section; decline section
Bounds: the system bound and the route-free bound (draft proofs, being rewritten); route slack in the bound (through corridor +45% under crane cadence, dock stopping roads +4–8%)	established	Chapter 3 bound section; dual-price column of chapter 5's rate table
Literature check: transfer sources checked; public sources on floating docks and giant blocks; Tao et al. 2017 compared in full	established	References; the introduction's comparison; floating-dock parameters

Source: this study's progress and results as of 10 Oct 2026 (each checked).

Running and queued: less conservative rules first, then retrofits and floating docks; each names the figure or section it produces

ITEM	STATUS	OUTPUT (WHERE IN THE THESIS)
Less conservative deadlock-free rules (a banker's-algorithm check plus step-wise feasibility search, or claims cut at the next safe holding point, the safe-orchestration theorem), with the refusal share. First step: the Okpo map with floating docks and branch roads, direction locks at stop U-turns, reservation waits split into crane / road	running	Chapter 4 rule comparison; re-check of the lower edge, the decline and the cost of safe orchestration
Retrofit simulations: three-resource dock network (ρ^* against dock-road slack); upgrades; block-length curve; door granularity	queued · after the rules	The proposition's threshold; the first road–vehicle exchange-rate table (chapter 5)
Floating docks, holding areas: floating-dock scenario; finite holding area S and $K_0(S)$	queued · after the retrofits	The decline's $K_0(S)$ curve and “how many vehicles a holding area is worth”; the vehicle-type hypothesis's floating-dock case

To decide once the rules are compared: the wording of the dock-mouth condition (readings on slide 8); the new numbers of the network-bound hypothesis (vehicles needed at yard-wide volume); the main basis (16 h or 24 h working day, daily mean or peak).

Risk	Safe release is conservative (2/3 of its waits start with a refusal), so its thresholds and decline numbers will move under a looser rule; the split's 200 m and 25 m are scenario settings.
Mitigation	The looser rule enters through the same runs; the result leads with the qualitative relations, numbers carrying rule and K ; both granularities are reported and the retrofit block-length sweep turns 25 m into a curve.

Sources: this study's plans as of 10 Oct 2026; risks and how to strengthen them, from the physical-rule runs. These three items have no results yet; they are listed as plans.

Target journals: the dock mouth goes into paper one (chapters 2–3); B-class first, RESS (A) as the stretch

First choice (B)

Transportmetrica A: Transport Science class B

Ocean Engineering class B

The capacity interval, p thresholds and waiting position are transport science; the dock mouth and erection are shipyard mechanisms.

Stretch (A)

Reliability Engineering & System Safety class A

Condition: the proposition is proved as a stability / resilience result, “blocking-induced starvation, instability and recovery”. This sub-study fits it best: crane starvation observes the instability; off-road waiting and splitting are the levers of recovery.

Alternatives (B)

IEEE Transactions on Automation Science and Engineering class B
Applied Mathematical Modelling class B

An automation-systems framing of safe release and sectioned control; a modelling framing of the bound and the three-resource network.

Supplementary (C)

Computers & Industrial Engineering class C
Journal of Ship Production and Design class C

Engineering outputs: design rules on holding bays, sectioned dock roads and floating docks.

B-class first; A-class as the stretch, condition on its card. The school requires at least one journal paper of class B or above; the segmentation paper is already published in Ocean Engineering (B), and the fleet-sizing paper targets Ocean Engineering.

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26. 中集来福士 CIMC Raffles (2012-07-11). 海工基地项目环境影响评价第二次信息公告 (2 × 370 t gantry, 20,000 t bridge crane).

To close: the dock-mouth section answers “where to wait, how to split the road, and how many vehicles wait on it”

1

The crane is in the model: erections at crane cadence, in- and out-of-span from the map, and one parameter p places a yard on the curve.

2

The mechanism is read (10 seeds): off-road waiting does not fall; rules that wait on the road, safe release without teleporting included, fall with p , with the threshold moving left with K and right with splitting; cranes can starve at nominal $p < 1$.

3

Next: a less conservative rule re-checks the lower edge and the decline; the three-resource network gives the threshold p^* for the proposition; together they become chapter 3's dock section and chapter 5's design rules.

[← Back to the thesis briefing](#)[Capacity interval →](#)[Traffic orchestration →](#)[Fleet-network co-design →](#)[Fleet-sizing paper: coupled carriage →](#)

Thank you.